TrafficFlowBench 2026
Method, Analysis and Results

Musa Peker (single-member team)
Ozveri R&D Center, Ozdilek Home Textile Industry and Trade Inc., Bursa, Türkiye

2026 IEEE Big Data Cup - final package for the selected submissions **s146** and **s148**

Summary

The selected finals are **s146** (public score 0.88588, private 0.87505) and **s148** (0.88481 and 0.87393). Task 1 is a blend of six convolutional inpainting networks and a LightGBM ensemble, with a dedicated reconstruction for the rows that the release blanks around every Task 2 window. Task 2 uses a public onset model and, for ongoing windows, a probability mix of two public pipelines and eight models of our own. Task 4 solves the documented prior-anchored least-squares problem to its exact optimum. The finding that shaped the second half of the competition is that the scored months are generated separately from the training months, so statistics learned from the training period mislead there. Removing the training-period priors, reading the released row at the origin and recomputing the priors from each window's own past (Section 2) added 0.0042 in three steps, of which the rolling priors contributed 0.0010.

	s146 (final)	s148 (final, hedge)
public score	**0.88588**	0.88481
private score	**0.87505**	0.87393
Task 1 networks	six networks fine-tuned to read ineligible values (s88)	the original six networks (as in s79c)
Task 1 rows inside the Task 2 blackouts	gated analogs, otherwise mean of 12 gap-aware networks	mean of 11 gap-aware networks, no analogs
Task 2, Task 4	identical	identical

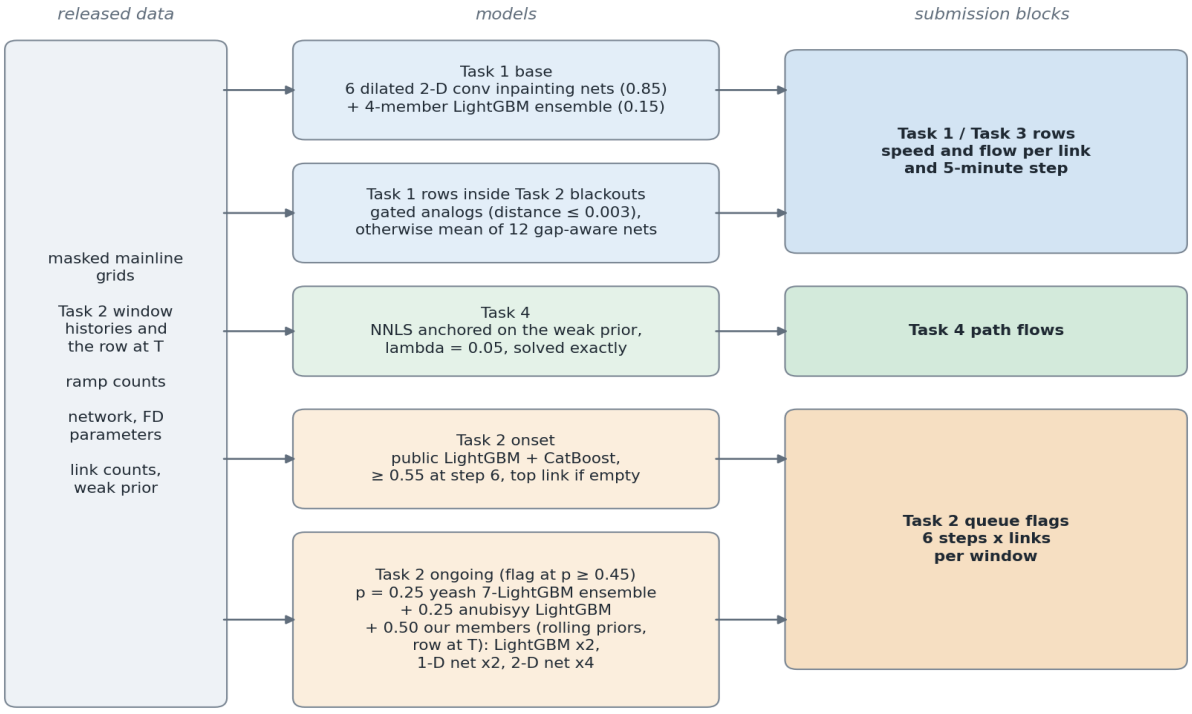


Figure 1. Data flow of s146. The hedge s148 uses the same Task 2 and Task 4 and an earlier Task 1 configuration (Section 6.1).

1. Evaluation and where the score comes from

The composite is $0.35 \cdot S_{state} + 0.30 \cdot S_{queue} + 0.15 \cdot S_{physics} + 0.20 \cdot S_{ODME}$ [1], where $S_{physics} = S_{FD}/3 + 2 \cdot S_{LWR}/3$ is evaluated on the Task 1 rows and every component is averaged over the two directions of a corridor and then over the five corridor families. The organizers publish the component scores of a naive baseline and of a perfect answer. With these anchors, a submission that changes a single task turns the public score into one equation in that task's component, which is how we estimated the values in Figure 2 and decided where to spend effort. S_{state}, $S_{physics}$ and S_{ODME} come from such probe submissions, and S_{queue} follows as (score $- 0.35 \cdot S_{state} - 0.15 \cdot S_{physics} - 0.20 \cdot S_{ODME}$) / 0.30.

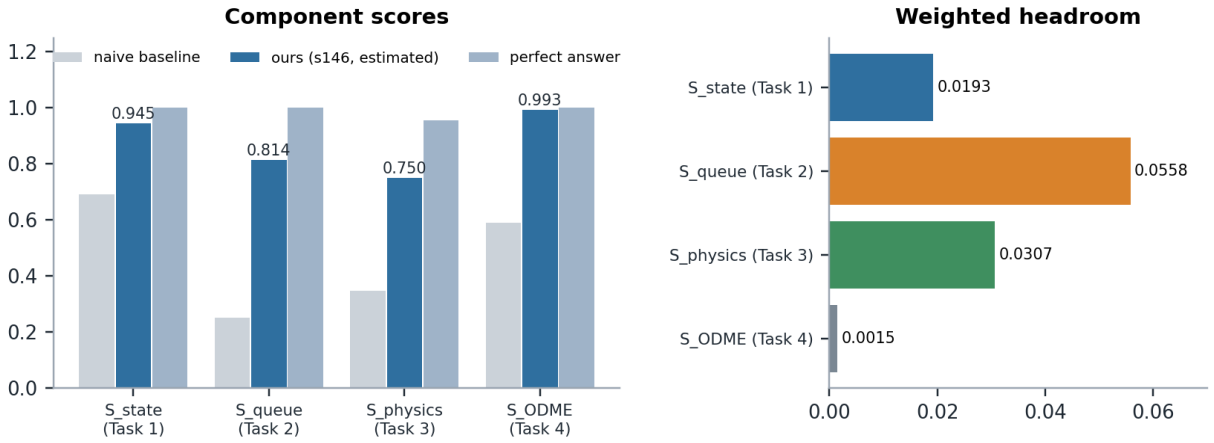


Figure 2. Estimated component scores of s146 next to the published naive and perfect references (left), and the weighted headroom of each component (right). Even an exact answer scores $S_{physics} = 0.954$, because conservation is checked against noisy observations.

Three properties of the metric mattered in practice:

- Task 2 counts predictions on every horizon cell but truth only on the eligible ones (about 76.5%). Eligibility cannot be predicted from data available at the origin (a LightGBM classifier trained on 27.8M cells reached an AUC of 0.54), so a positive forecast on an ineligible cell is an unavoidable false positive.
- S_{state} is linear in RMSE: $0.54(1 - RMSE_v/25) + 0.46(1 - RMSE_q/600)$ per regime, with flow measured per lane. S_{LWR} scores the change of accumulation $N = (q/v) \cdot L$ between consecutive steps, as in the LWR conservation law [2, 3], in which a flow error weighs about 2.5 times as much as a speed error.
- A Task 2 forecast at origin T may use traffic data stamped at or before T: the window history, the released row at T, earlier days of the same split and, for private windows, the validation month. Train and the validation month may be used for fitting and selection; private observations stamped after T may not be used for anything, including comparisons. At award review the organizers re-run Task 2 with all later rows removed and on a newly generated month.

2. Train and scored months come from different generations

Our first Task 2 models used priors estimated on the training months: queue rates per link and per time of day, and the average horizon pattern of onset and ongoing windows. On held-out training days they were among the strongest features; on the leaderboard they were not. Figure 3 shows the reason. On D7_I405_N the share of queued link-steps falls from 9.7% in training to 8.0% in the validation month. On D12_I5_S the queues form in different places: in the validation month the per-link queue rate correlates only 0.62 with training.

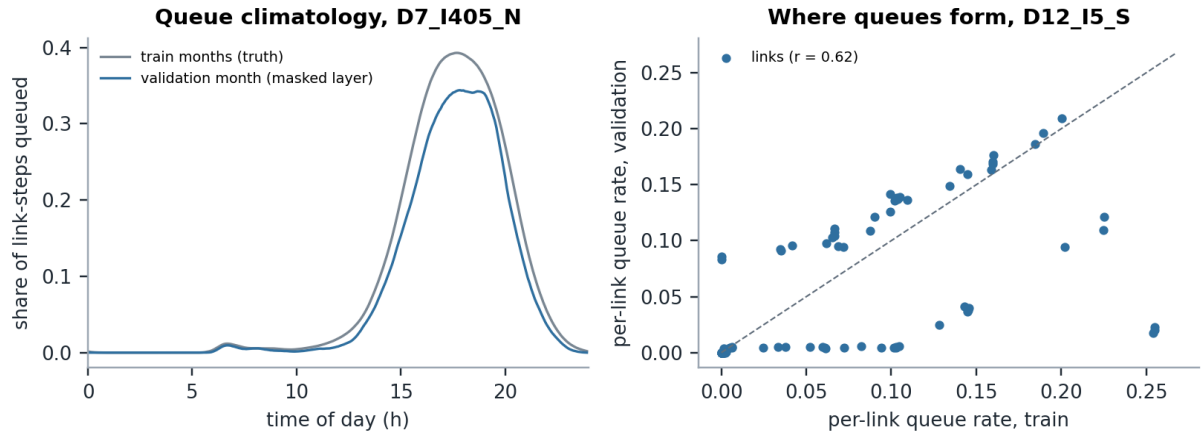


Figure 3. Queue climatology of the training months (ground truth) and of the validation month (its released masked layer). Left: share of queued link-steps by time of day. Right: per-link queue rate, training against validation.

A holdout cut from the last training days belongs to the training generation and does not show this effect (Figure 4, left). We removed the dependence in three leaderboard-tested steps: models without priors (s105 to s111, +0.0018 over s101); the same models also reading the released row at the origin (s113, +0.0014); and **rolling same-split priors** (s118, +0.0010). A rolling prior is recomputed for every window from the 14 days before the window's day, on the Task 1 masked layer of the window's own split; for a private window the validation month supplies the earlier days. On the first validation day, which has no earlier day in its split, the last 14 training days are used.

To measure the shift locally we built a **pseudo-window harness**: Task 2 windows selected with the organizers' rules inside the validation month's masked layer (229 ongoing and 200 onset windows, independent of the 80 scored ones) and scored on the visible cells. It ranks seven of our submissions in their leaderboard order (Figure 4, right). It disagreed with the public board in three other cases, which involved the onset condition, the public models or the decision threshold. We attributed this to two properties of the harness: an onset defined on visible cells alone admits queues that were already present in hidden cells, and the pseudo histories are sparser than real ones (58% against 83% of cells), which penalizes models trained on dense inputs. We therefore used the harness only to compare our own ongoing members. For onset this was too cautious: the private leaderboard sided with the harness (Section 6.2).

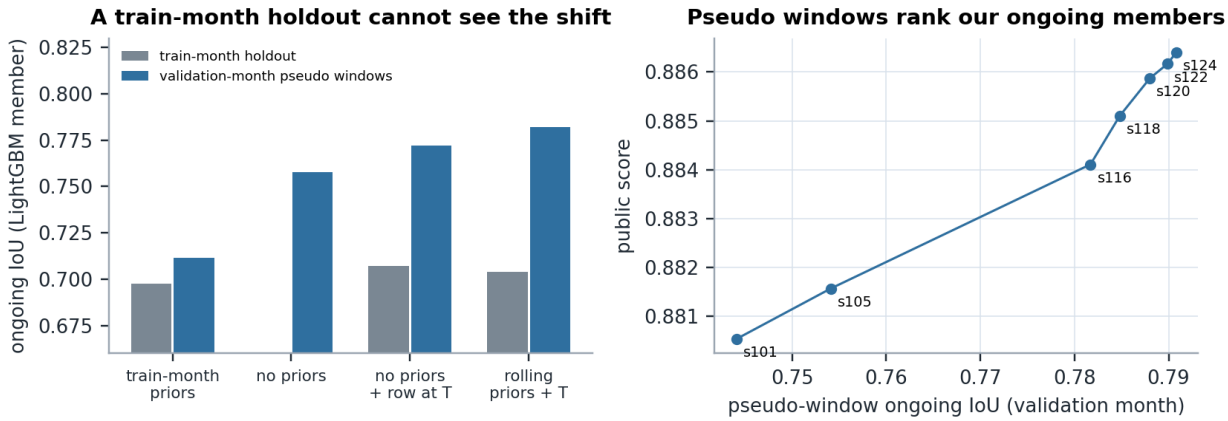


Figure 4. Left: ongoing IoU of one LightGBM recipe with training-month priors, without priors, without priors but with the row at T, and with rolling priors and the row at T, on a training-month holdout (the variant without priors was not run there) and on validation-month pseudo windows. Right: pseudo-window IoU of seven submissions against their public score.

3. Task 1: traffic-state reconstruction

3.1 Tree ensemble

Four LightGBM [4] models predict speed, flow per lane and density for every masked cell. Their features are the nearest observations of the same link before and after the cell, the nearest observed links up- and downstream, time-of-day profiles computed from the observed cells of the same split, rolling means, occupancy and ramp-count balance terms. The models differ in scope: one per corridor, one for all corridors, one per corridor family, and a shallow extremely randomized variant. Per-target least-squares weights, fitted on the last 45 training days, combine them.

3.2 Convolutional inpainting networks

Most masked cells are isolated and surrounded by observations, so a convolution over the time × link grid sees far more context than hand-made features. Each network reads nine channels (speed/v_f, flow/capacity, occupancy, observed mask, share observed, v_f, lanes, capacity, time of day) through GroupNorm [5] and GELU residual blocks with dilated convolutions [6] up to dilation 16 or 32, at 128 channels (one network uses 192). Training is self-supervised on the training months: 20%, 30% or 50% of the observed cells of a crop are hidden, matching the three masking regimes of the release, and the loss on the hidden cells is the metric's own weighting, $0.54((v_{pred} - v)/25)^2 + 0.46((q_{pred} - q)/600)^2$ per lane. The six networks differ in seed, width, depth of dilation and crop size; on the leaderboard, structural differences between members helped and further seeds of the same structure did not.

About 7% of the released cells hold a value with pct_observed below 75. These cells are never targets but are nearly as informative as eligible ones; feeding them to the networks with their pct value, first at inference (s87) and then also in fine-tuning (s88), gained 0.0006. The Task 1 output is 0.85 × the mean of the six networks + 0.15 × the tree ensemble. The weight was set on $0.35 \cdot S_{state} + 0.15 \cdot S_{physics}$, because most of what the networks add arrives through S_{LWR}.

3.3 Rows inside the Task 2 blackouts

Around every scored Task 2 window the release blanks the whole corridor for the 30-minute horizon and the 60 minutes after it, 18 steps in total, while the Task 1 template keeps its targets there: 128,465 rows, 1.9% of all scored Task 1 rows, all of them at queue onsets or ongoing queues. Networks trained on scattered gaps filled these blocks with free-flow values, with an in-gap RMSE of 12.6 km/h and 300 vph per lane against 1.3 km/h and 36 vph elsewhere. This explained why the leaderboard implied a Task 1 error about 1.55 times larger than our holdout.

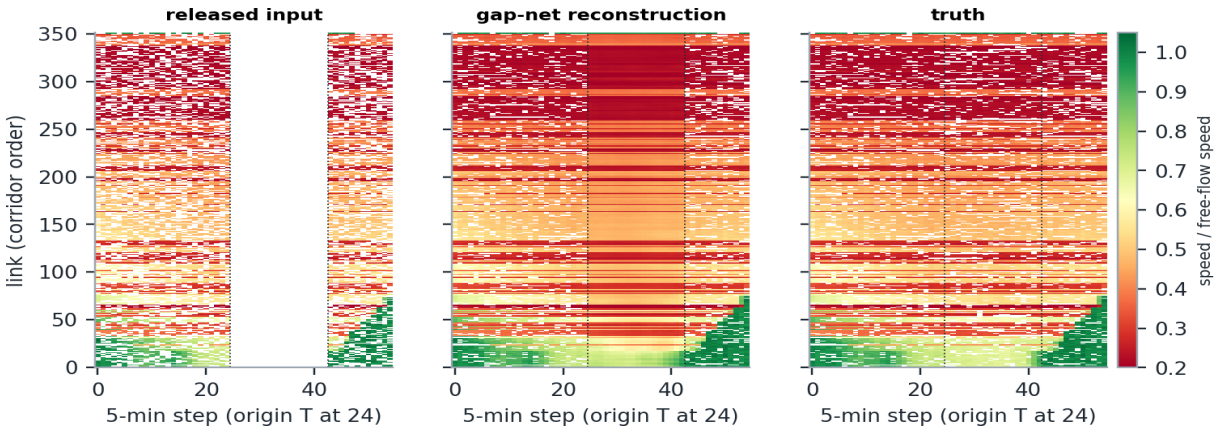


Figure 5. An organizer-style blackout on a held-out training day (D7_I405_N, ongoing queue): released input, reconstruction by the gap-aware networks, and ground truth. Colour is speed relative to free-flow speed; the dotted lines bound the 18 blanked steps.

Two reconstructions are combined. **Gap-aware networks** are the Task 1 networks fine-tuned with synthetic 18-step full-corridor blackouts in half of the training crops, later placed at queue onsets and ongoing queues as the organizers place theirs; the set grew from seven networks (s73) to eleven (s78) and twelve (s83, one trained from scratch), and its members were later fine-tuned to read the ineligible values as well (s89, s95). **Analogs** describe a gap by the corridor's speed and flow ratios over the six steps before and after it, and average the true trajectories of the ten closest moments at the same time of day, searched in the training months or in the scored month's own

other days, whichever matches better. Analogs on every gap lost on the leaderboard (s68, –0.0011), because the separately generated months match the training library much less closely than the training months match each other. They are therefore used only where the match distance is at most 0.003, and the networks fill the remaining gaps. These steps added 0.0081 from s55 to s73 and a further 0.0002 up to s86, when the analog library was extended to the scored month itself.

3.4 Task 3: physical consistency

Task 3 is computed from the Task 1 rows, so every Task 1 improvement also raised it. S_{FD} on our rows is 0.991, slightly above the 0.986 of the ground truth itself, which leaves S_{LWR} (about 0.63) as the open part. Examining the generator showed that free-flow speed is v_f plus a per-timestamp shock shared by all links of a corridor (about a third of the noise variance) plus independent noise per link. The networks follow the shared shock (correlation 0.88), and outside the blackouts their speed error equals the independent noise. Density smoothing, projection onto the fundamental diagram, an S_{LWR} loss term, a separate density head, test-time augmentation and a cell transmission model for the blackouts were tested and did not help.

4. Task 2: online queue forecasting

Each window requires a binary queue flag (speed ≤ $0.6 \cdot v_f$) for every link at the six 5-minute steps after the origin T. In onset windows the corridor is free of queues during the history and a queue first appears at step 6; in ongoing windows a queue is present at T and in the horizon, and the IoU of the persistence forecast (the queue flags at T repeated over all six steps) with the truth is at most 0.9.

4.1 Onset

Onset probabilities are 0.5 × LightGBM + 0.5 × CatBoost [7] from the public anubisyy pipeline [8], which we retrained from its published code. Links with $p \geq 0.55$ are flagged at step 6 only, and a window without such a link receives its most probable link, since every scored window contains a queue. An onset window has only about three true cells, so a single changed cell moves the composite by roughly 0.0001, and every alternative onset model or decoding rule we submitted lost on the public board (Figure 9); Section 6.2 shows that several of them did better on the private month. Retraining the public onset models from their code reproduces the notebook's published labels on all but 4 of the 174,000 queue cells. The replica gives these cells probabilities of 0.51 and 0.56, close to the 0.55 threshold, and 0.43 and 0.45, which the original models must have placed above it. Each of the four windows has other links above the threshold, so the rule for empty windows is not involved; the difference comes from retraining, which is not bit-exact across machines and library versions. s146 and s148 carry the notebook's published labels on these four cells; they were copied from the notebook's output file, which the code package does not read (Section 9). The four cells are worth 0.0009 on the public board (s131 against s146).

4.2 Ongoing

The ongoing probability is $p = 0.25 \cdot p_{yeash} + 0.25 \cdot p_{anubisyy} + 0.50 \cdot p_{ours}$, flagged at $p \geq 0.45$. p_{yeash} is the mean of the seven-member LightGBM ensemble of the public notebook by yeashusemwal [9] and $p_{anubisyy}$ a public LightGBM [8], both retrained from their published code. p_{ours} is the mean of the three members below, each averaged over its seeds:

member	model	inputs	seeds
LightGBM	binary gradient boosting on (link, step) rows; 127 leaves, 500 rounds; 2,800 onset and 2,800 ongoing training windows per corridor	summaries of the 60-minute history (latest speed ratio, trend, minimum, queue flags, share observed) for the link and its neighbours, corridor summaries, link attributes, rolling priors, row at T	2
1-D network	dilated convolution along the corridor, 192 channels, dilations up to 64	history as channels per link, row at T, no priors	2
2-D network	dilated convolution over history step × link, 96 channels	history grid and row at T; two seeds without priors, two with rolling priors	4

The organizers confirmed that the released row at T may be used. It comes from the Task 1 masked layer, where about a third of its cells are blank, so during training 35% of that row's cells are hidden at random. The two families of models complement each other: dropping the public members lost 0.0014 (s126), and replacing the training-month weekday profile inside the yeash ensemble with a rolling one lowered its pseudo-window IoU from

0.754 to 0.709, because the ensemble was trained on dense profiles.

Our members were built up in tested steps: averaging two seeds of each network added 0.0004 (s116); the rolling priors in the LightGBM member added 0.0010 (s118); 2-D networks trained with rolling priors and a second seed of the LightGBM member added another 0.0013 (s120 to s124).

4.3 Remaining errors and the threshold

On validation-month pseudo windows, 77% of the ongoing false positives fall on visible but ineligible horizon cells (Figure 6, left). Since eligibility is unpredictable, these errors cannot be removed. On eligible cells there are almost exactly as many false positives as false negatives (705 and 711), so the mix is well calibrated there. The threshold 0.45 is the leaderboard optimum among the three values tested.

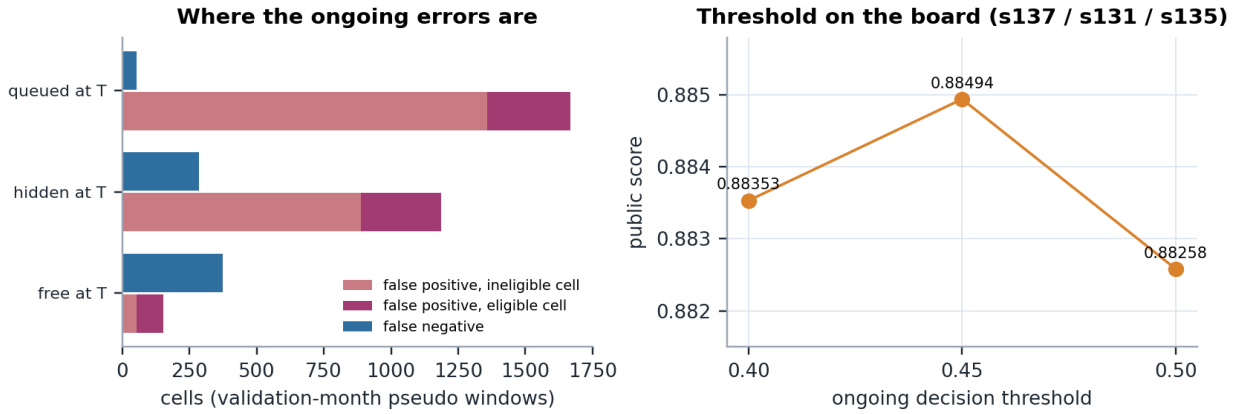


Figure 6. Left: false positives and false negatives of the s131 ongoing mix on validation-month pseudo windows, by the state of the link at the origin and by eligibility of the horizon cell. Right: public score for three ongoing thresholds.

4.4 Compliance with the online rule

All Task 2 inputs are stamped at or before T. We checked every fallback branch against the organizers' re-run with later rows removed and found one violation: on the first validation day the rolling priors fell back to the window's own day. After the fix the 16 windows of that day use the last 14 training days. No private window was affected. s146 is s124 with this fix and nothing else, so the fix cost 0.0005 of public score (0.88640 to 0.88588), all of it on the 16 corrected windows. s146 and s148 are produced by the corrected code.

No private-month observation stamped after a window's origin was used to fit, select or compare anything in s146 or s148. All Task 2 choices were made on the training months and the validation month (public board and validation-month pseudo windows), and the recipe was fixed with s124 on 2026-10-18. Afterwards we also ran the pseudo-window harness on the private month while testing further ideas; none of those tests led to a change in the final submissions, and their results are not used in this report.

5. Task 4: origin-destination estimation

For every corridor and split we solve $\min_{f \geq 0} \tfrac{1}{2}||Af - c||^2 + (\lambda/2)||f - b||^2$ with $\lambda = 0.05$, where A is the released path-link incidence matrix, c the released link counts and b the released weak prior of that split. No model is trained. Anchoring on the released weak prior instead of a reconstructed OD matrix gave the largest gain of the first week (+0.074 on the board before the data rebuild). Projected FISTA [10] stopped up to 3.5% (in L1 distance) short of the optimum on the large corridors, so L-BFGS-B [11] continues from its solution (s90, +0.00025). S_{ODME} is about 0.99.

6. Results

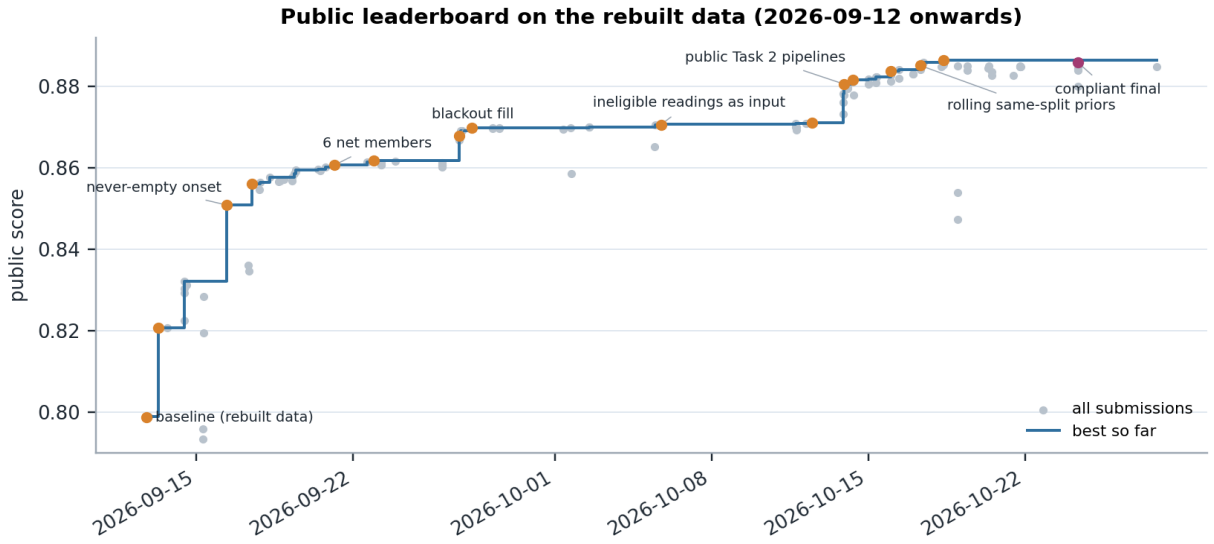


Figure 7. Public score of every submission on the rebuilt data (grey), the best score to date (blue) and the milestones of Figure 8 (orange).

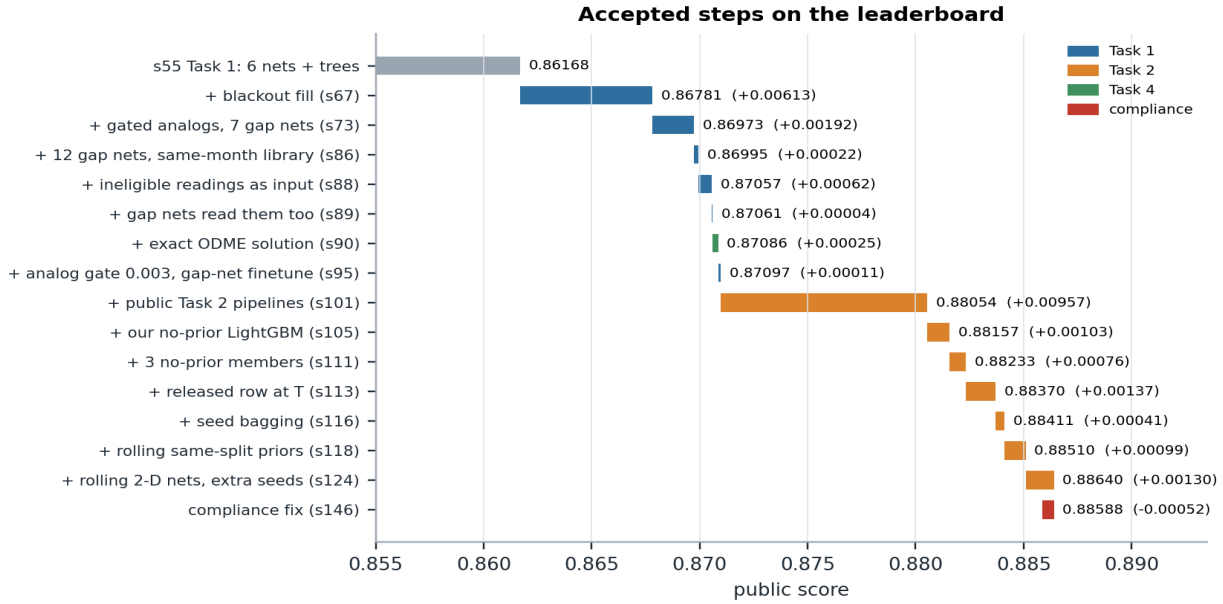


Figure 8. Accepted steps from the stable Task 1 base to the final submission. Each step was a separate leaderboard test; most changed one thing, s86, s95 and s124 two closely related things; the compliance fix of Section 4.4 is the only step that lowered the score (–0.0005).

submission	public	description
s146 (final)	**0.88588**	corrected code, public onset labels on the 4 replica cells
s148 (final, hedge)	0.88481	Task 1 of s79c (no analogs, no ineligible readings), Task 2 and Task 4 of s146
s131	0.88494	output of the code package for s146 (differs on the 4 replica cells)
s124	0.88640	highest public score, produced before the compliance fix; not selected

On the private leaderboard the two finals scored 0.87505 (s146) and 0.87393 (s148); the team's private score is that of s146.

6.1 Leaderboard noise and transfer to the private month

The public score is computed on the validation month, with 80 Task 2 windows. A single onset cell is worth about 0.0001, so differences below roughly 0.0005 are within the noise of the public board, and 135 submissions give a real risk of fitting it. We limited that risk in four ways. First, this noise estimate concerns Task 2, which is scored on 80 windows; Task 1 and Task 4 are scored on 6.7 million and 70,708 rows, where even the small steps of Figure 8 (s86, s89, s90, s95) are measured with little noise, and the Task 2 steps that changed the model structure are worth 0.0008 to 0.0096 each. Second, the main changes to our ongoing members were also measured on 229 ongoing pseudo windows of the validation month, against 40 ongoing windows on the board; the gains from removing training priors, adding the row at T and using rolling priors appear there in the same order and with larger margins (Figure 4, left). Third, few parameters were chosen on the board, and each is named here: the ongoing threshold 0.45 (its neighbours lost 0.0014 and 0.0024, Figure 6), the ongoing mix weights 0.25 / 0.25 / 0.50 (s111) and the analog gate, first set to the typical match distance within the training months (0.002) and raised to 0.003 on the board (s95). The Task 1 blend 0.85 / 0.15 was set on the networks' holdout, the onset threshold 0.55 is the public pipeline's own, and $\lambda = 0.05$ is the documented value of the ODME objective. Fourth, the small steps (seed averages and additional members of an existing kind) only average models and do not tune anything to the validation windows.

Choice of the hedge. s146 and s148 share Task 2 and Task 4 and differ in Task 1. Three Task 1 steps of s146 depend on properties of the scored month: the analogs need the edges of a gap to match the training months or the month's other days, the same-month library uses the month itself, and the ineligible readings are a feature of how each month was masked. s148 uses the Task 1 of s79c (our earlier hedge s79, which had no analogs, with the converged Task 4 solution of s90; c for converged), which contains none of these steps (the original six networks, and 11 gap-aware networks on every blackout row). Across eight pairs of submissions built this way (s113/s114 to s146/s148) the hedge scored 0.00107 to 0.00108 lower on the public board every time, the same difference as between their Task 1 configurations alone (s95 against s79c). The hedge therefore costs a known, small amount if those steps transfer to the private month, and protects the submission if they do not.

6.2 Private leaderboard

The private scores, published after the competition closed, allow several claims of this report to be checked. Across the 93 rebuilt-data submissions with a public score above 0.86, public and private scores correlate at 0.975 (rank correlation 0.94), and the private score is lower by 0.0082 on average (standard deviation 0.0022).

claim	public	private
Task 1 and Task 4 steps transfer (s55 to s95)	+0.0093	+0.0096
structural Task 2 steps transfer (s95 to s118)	+0.0141	+0.0106
small late steps are within the noise (s118 to s124)	+0.0013	–0.0001
the compliance fix and the replica cells leave the private score unchanged (s124, s131, s146)	0.88640 / 0.88494 / 0.88588	0.87505 for all three
cost of the hedge (s148 against s146)	–0.0011	–0.0011

One result went the other way. The three onset alternatives that lost on the public board all score higher on the private one: our rolling-prior onset heads (s127, 0.84733 public and 0.88784 private, +0.0128 over s124), our earlier onset models (s106, +0.0021 over s105) and the yeash onset heads (s138, +0.0013 over s131). The validation-month pseudo windows had pointed the same way, with an onset IoU of 0.75 against 0.66 for the public model, but we followed the public board, whose 40 onset windows were evidently unrepresentative for these models. An onset decision taken on 40 windows of about three true cells each needed a larger sample than the public board could give.

7. What did not work

Of the 135 submissions on the rebuilt data, Figure 9 lists the instructive failures of the final phase. Two results deserve a comment beyond the figure.

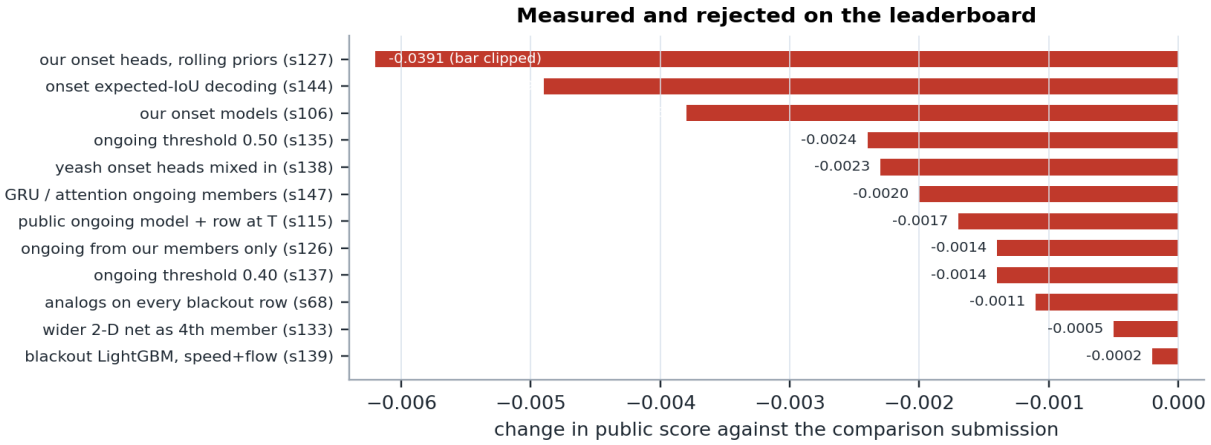


Figure 9. Leaderboard tests that lowered the public score, relative to the submission each one modified; Section 6.2 compares the onset tests with the private board.

- **Better local reconstruction of the blackouts did not transfer.** A LightGBM model for the blackout rows reduced the in-gap RMSE by 11% for speed and 21% for flow on held-out training days and left the public score unchanged (s139, s140). A version trained on synthetic blackouts in the scored months' own masked layer was also neutral (s141, s142).
- **Our ongoing members saturated at three structures.** A wider 2-D network (s133, s134), GRU and attention members (s147), a speed-regression LightGBM and a LightGBM with a three-hour context did not improve the mix.

8. Validation

- Task 1: the networks' own seven-day holdout, scored with the competition's formulas for S_{state}, S_{FD} and S_{LWR} and aggregated by panel and regime.
- Blackouts: organizer-style windows inserted into training days, scored only on days the networks did not see.
- Task 2: windows from the last 60 training days, which no model was trained on, scored with predictions counted on every cell; and the validation-month pseudo-window harness of Section 2 for comparing our ongoing members.
- Leaderboard: one change, or two closely related ones, per submission, with the expected effect written down before the result.

9. Reproducibility

The code package contains a README, the list of requirements, the source code, the archived model weights with their md5 checksums, TRAINING.md (command, environment, seed and held-out days of every model) and a single entry script:

```
python reproduce.py --data <release> --work <dir> --final s131   # s146
python reproduce.py --data <release> --work <dir> --final s132   # s148
```

The run names are those of the code's own outputs: s131 and s132 are s146 and s148 without the four published onset labels of Section 4.1. The --final option selects the configuration; s132 combines the Task 1 configuration of s79c with the Task 2 and Task 4 of s131. The times below are for inference from the archived weights; training times are listed per model in TRAINING.md.

stage	content	inference time*
caches	time × link grids from the parquet release	50 min
task4	prior-anchored least squares, FISTA followed by L-BFGS-B	2 min
task2	our eight members, public onset and ongoing models, probability mix	50 min
trees, nets	Task 1 tree ensemble; six networks	120 min

stage	content	inference time*
gap networks, gap fill	twelve gap-aware networks, gated analogs	25 min
package	merge with sample_submission.csv, checks, zip	3 min

* Intel i7-12700H, 32 GB RAM, NVIDIA RTX 3070 Ti Laptop GPU (8 GB), Windows 11, Python 3.11, PyTorch 2.6 with CUDA 12.4, LightGBM 4.6, CatBoost.

Starting from an empty working directory, the package reproduces s146 and s148 exactly on all Task 1, Task 3 and Task 4 rows and on all but the 4 replica cells of the 174,000 queue cells (Section 4.1). The effect of those cells is 0.0009 of public score, well within the 1% tolerance. No external data, pretrained model or paid service is used, and no file from before the data rebuild of 2026-09-11 is read. The two public Task 2 pipelines are retrained from their published code. The blackout reconstruction belongs to Task 1, which is an offline task, and uses released observations on both sides of each gap. Two early diagnostic submissions (s24, s54) read the horizon eligibility flags; they are not selected.

10. Conclusions

Most of our late progress came from analysing how the scored data were built rather than from new models: the blackouts around Task 2 windows explained the Task 1 gap, and the separately generated scored months explained why training-period priors failed. Recomputing priors from each window's own past is a simple, causal remedy that also respects the online rule. The private leaderboard confirmed these structural steps and showed that the public board was a poor judge of onset changes: a local harness built from many more validation-month windows was, in the end, the better guide there.

References

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